

Project Initialization and Planning Phase

Date	08 October 2026
Team ID	SWTID-2026-2386
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

EduGenie is a lightweight, GenAI-powered learning assistant. It turns a student's question, topic or study notes into a clear answer, a structured explanation, a summary, a quiz or a learning path.

Project Overview

Objective	Help students understand, revise and test academic topics using Google Gemini, in a simple web page focused on learning
Scope	Five features (question answering with follow-ups, concept explanation, summarization, quiz generation, learning path); no accounts, no database, no file uploads

Problem Statement

Description	Students find explanations too complex, notes too long to revise and practice questions hard to get for their own topic
Impact	Solving this saves study time, improves understanding and helps students check their own progress

Proposed Solution

Approach	A FastAPI backend validates the student's input, a feature module builds a task-specific prompt, and one shared Gemini service sends it (with an automatic model fallback) and returns the result. The quiz reply is requested as JSON, cleaned, parsed and validated before it reaches the page. The API key stays on the server
Key Features	- Ask a question, with optional follow-ups - Structured concept explanations with an example - Summaries of pasted study notes - Three-question quiz with instant feedback and corrected answers - Learning path from beginner to advanced - Friendly error messages and key protection

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Any laptop, no GPU (inference runs in Google's cloud)
Memory	RAM specifications	4 GB minimum; the server itself is small
Storage	Disk space for data, models and logs	Under 1 GB; no model download and no database
Software		

Resource Type	Description	Specification / Allocation
Frameworks	Python frameworks	FastAPI, Uvicorn, Jinja2
Libraries	Additional libraries	google-genai, pydantic, python-dotenv, pytest
Development Environment	IDE	VS Code, Git, GitHub
Data		
Data	Source, size, format	No training dataset; the student's text at run time (JSON) and Gemini responses (text or JSON). Nothing is stored